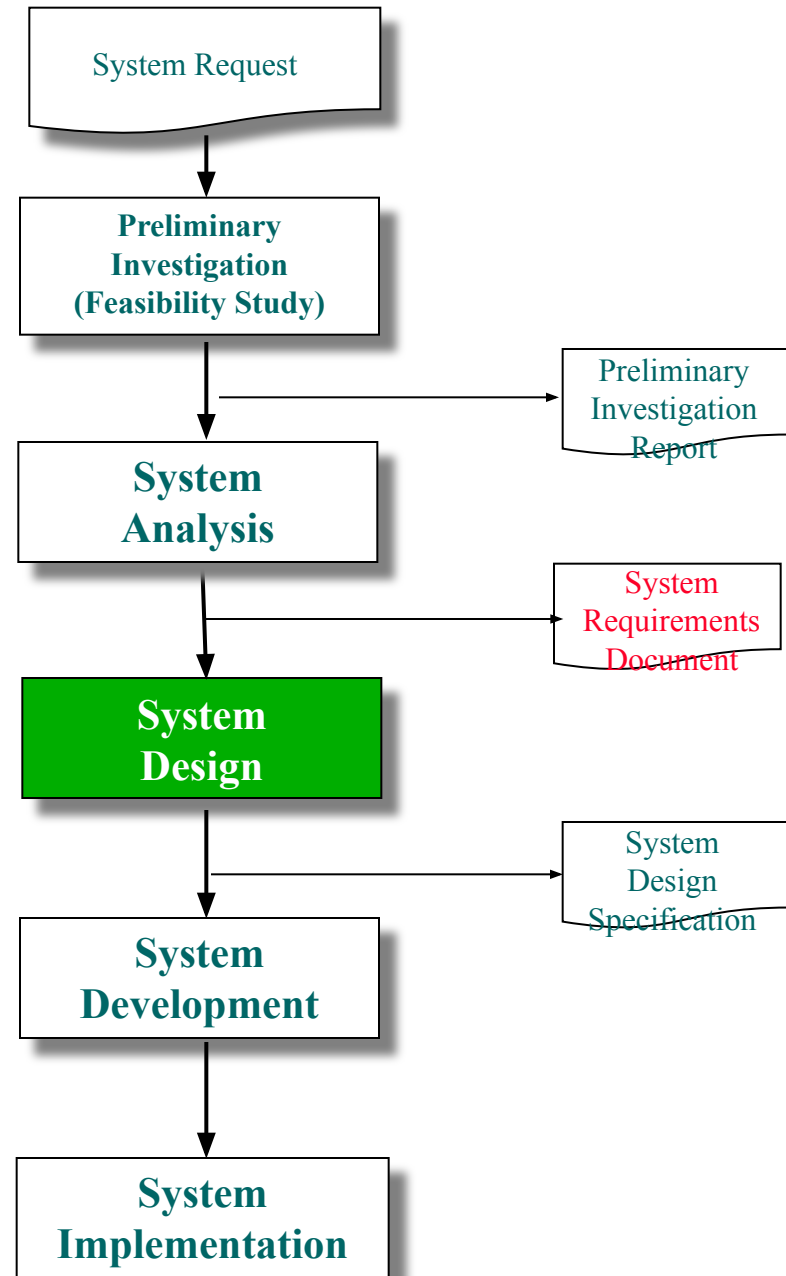


Chapter 7

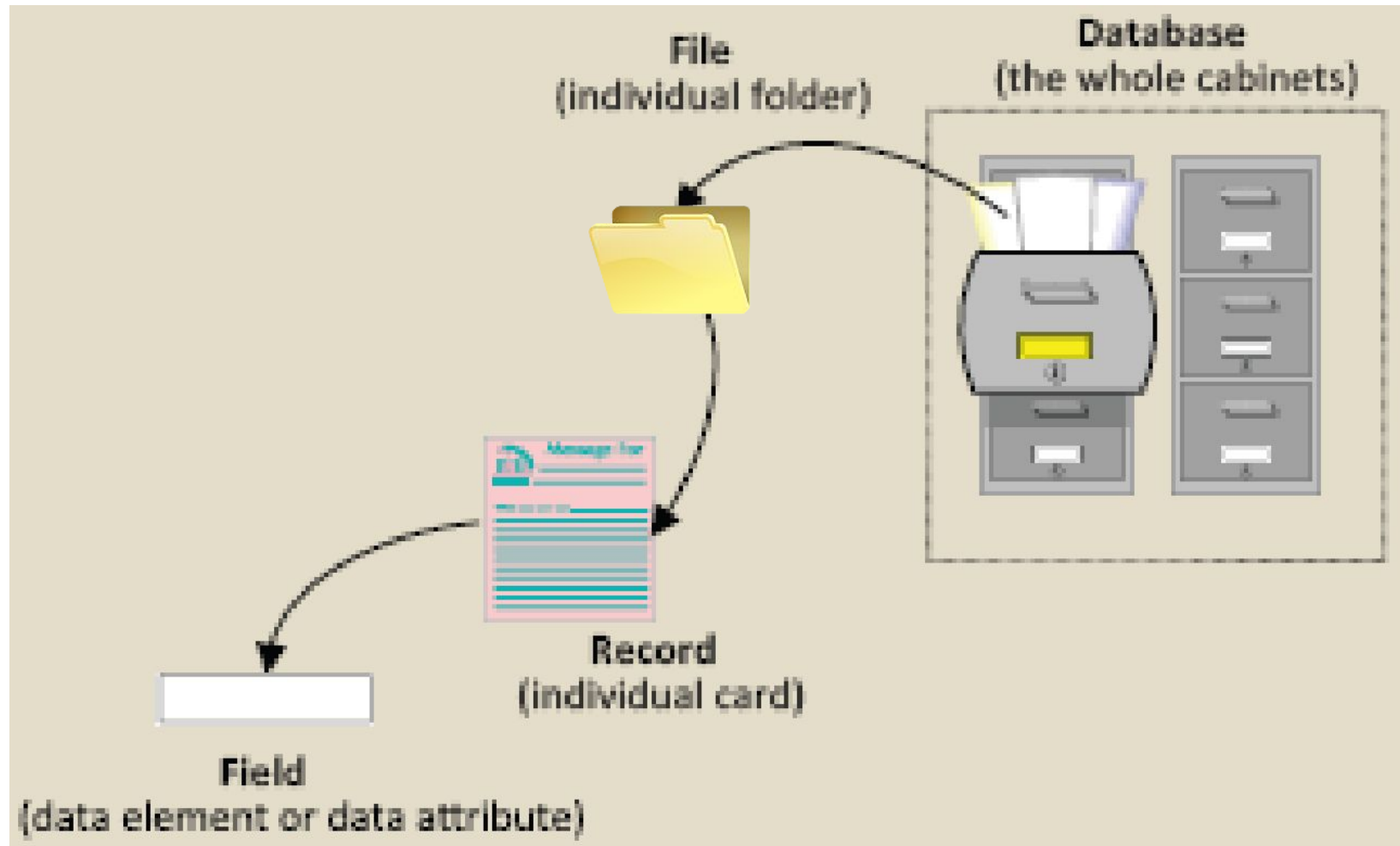
Database Design

SDLC



Database Design

Database Concept (Manual)



Database Concept (Computer)

- **Database** – consists of a number of files/tables.
- **Table (File)** – collection of related data records.
- **Record** – collection of related items of data fields.
- **Field** – contains one logical unit of data characters.



Key Field

- **Meaning** – a field used to identify and differentiate one **record** from other records in a table.

Mater Files	Key Fields
Payroll	Employee No./ ID
Customer	Customer No. / ID
Stock	Stock Code

Recommended:

Human – Use No. / ID

Transaction – Use No.

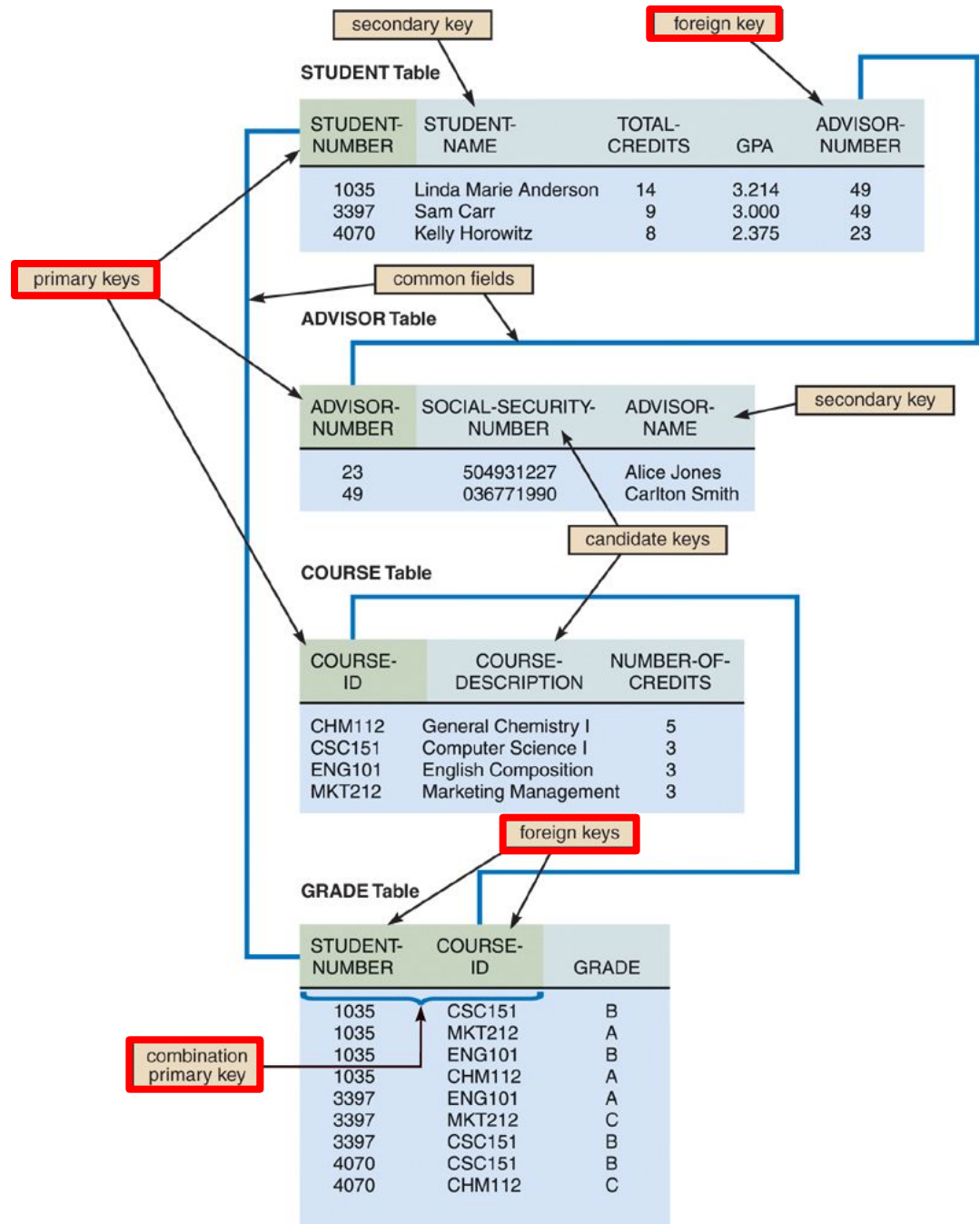
Item – Use No. / Code

Transaction Files	Key Fields
Invoice	Invoice No.
Purchase Order	PO No.
Payment	Payment No.

Types of File/Table

- **Transaction File** – contains transaction records captured from business transactions which changes frequently with the latest transaction data eg. **Purchase Order file** contains details of a PO such as PO number, PO date. Examples: PO, SO, Invoice, Payment, RFQ, Quotation etc.
- **Master File** - contains relatively more permanent records in a table eg. a **customer file** will contain details of a customer such as customer ID, Name and Contact Address. Examples: Stock, Supplier, Student, Customer, Branches etc.
- **Organisation File** – contains organization data used to help set up the organization structure eg **Company file** contains details of a Company such as Company No. Name. Address.

Types of Keys

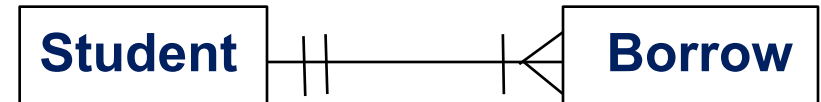


1. Primary key

- **Definition** - a field or a combination of fields that uniquely identifies a particular record of an entity. A primary key must be unique.

- **Examples**

- Student Table – **Student ID**
- Borrow Table – **Borrow No.**



Student Table

Student ID	Name	Phone		
WM23D01	Tan Ah Meng	0335454884		
WM22B12	Lilian Wong	0445982378		

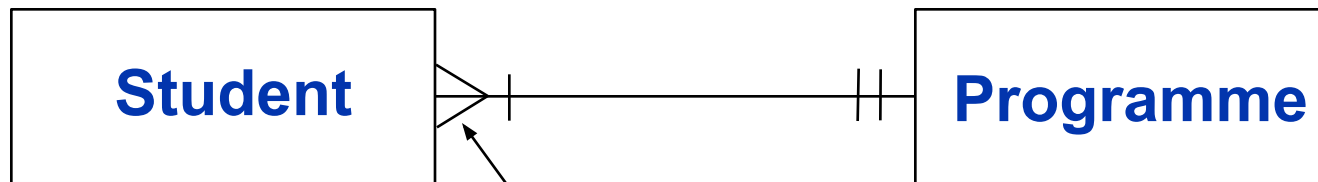
Borrow Table

Borrow No.	Date	Student ID		
000000017	24-12-23	WM22B12		
000000018	24-12-23	WM23D01		

Try fill up
other data
fields

2. Foreign Key

- **Definition** – A field in one table that must match a primary key of another table and borrowed.
- **Purpose** – To establish the relationship between two tables (entities)

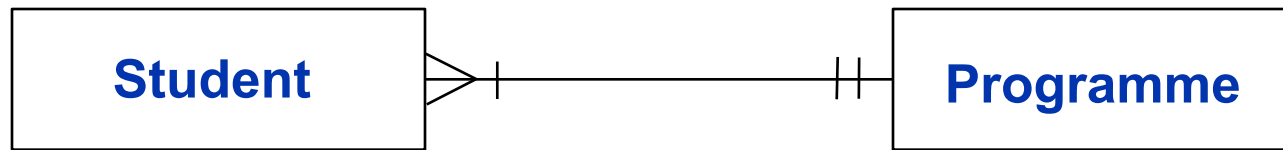


The '**Many**' symbol indicates that the **Foreign key** 'Programme Code' exists in the **STUDENT** entity

STUDENT (Student Number, Name, Address, Email, Telephone, Programme Code*)

PROGRAMME (Programme Code, Programme Name, Type)

Foreign Key



Student Table

Student-Number	Name	Address	State	Telephone	Programme Code
1001	Tan Beng Hong	8, Jalan Bertuah 3, Taman Segar	SGR	78994332	BT0005
2214	Lim Chee Keong	14, Jalan 3/4G, Taman Desa	KUL	83463639	AC0002
1034	Ahmad Bin Ali	2A, Lorong 14/2, Wangsa Maju	KUL	32263840	BT0005
7435	John Dykes	13, Jalan Merlinja 3, Taman Rinting	PRK	63673764	SC0001

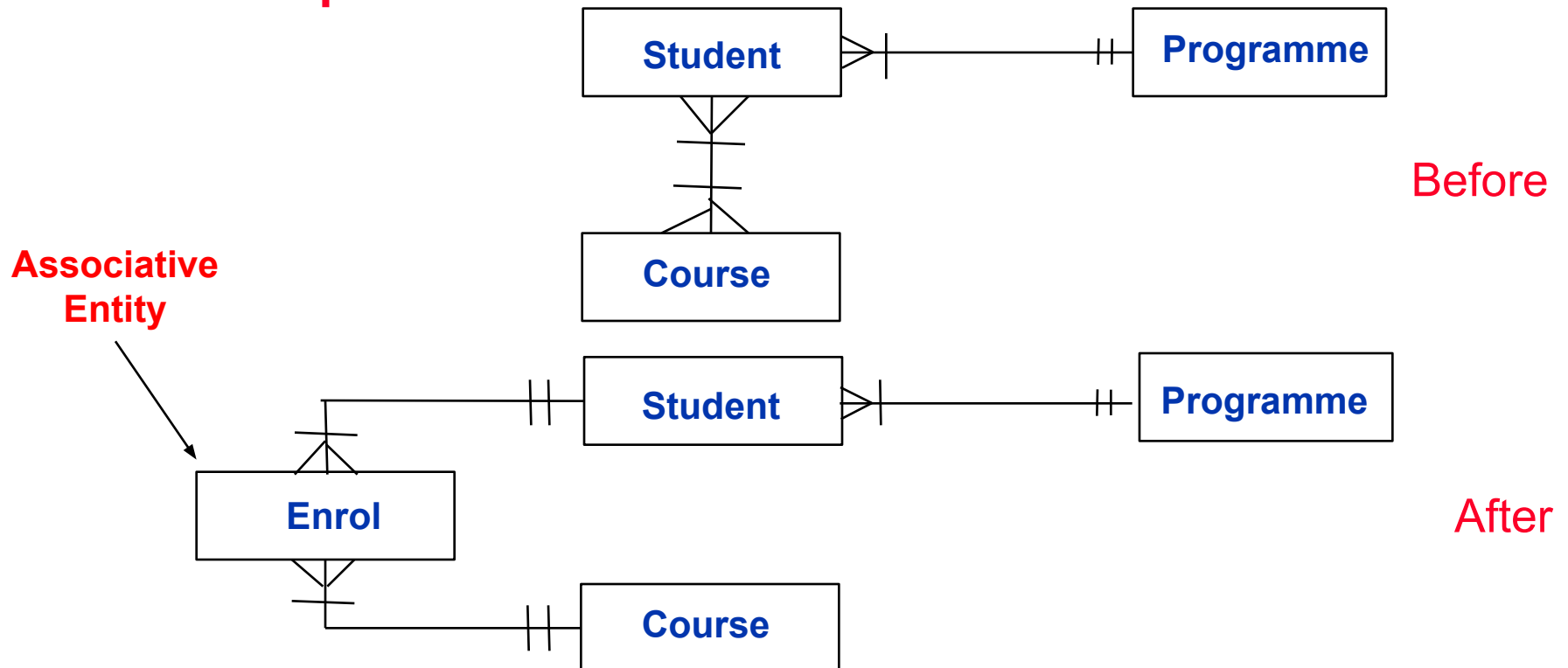
Programme Table

Programme Code	Programme Name	Faculty
SC0001	Degree in Biology	Sciences
BT0005	Degree in Engineering	Building Technology
AC0002	Degree in Accounting	Accountancy
IT0004	Degree in Compute Science	Information Technology
SC0002	Degree in Pure Physics	Sciences
MD0003	Degree in Medicine	Medicine

Foreign Key

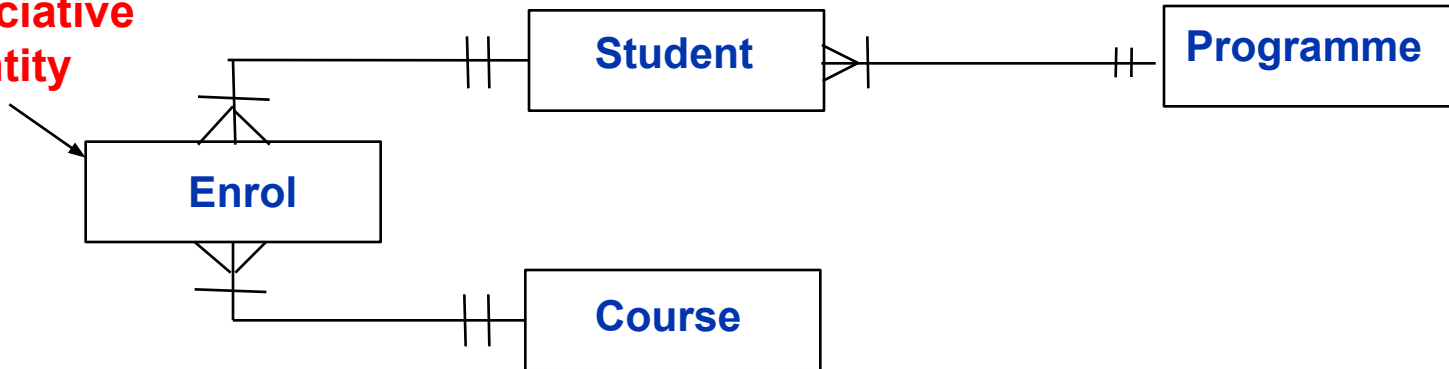
3. Compound Key

- **Definition** – created by combining the primary keys from two tables which have relations. Normally used in an associative entity'.
- **Purpose** – server as a primary key and used to link (connect) to other tables (ie foreign keys).
- **Example** : Student Number and Course ID



3. Compound Key

Associative
Entity



Enrol Table

Student No.	Course Code	Date		
WM12D07	AACS1304	22-1-24		
WM78B88	AACS3074	04-1-24		
WM64D11	AACS3311	18-1-24		
WM78B88	AACS1304	04-1-24		
WM64D11	AACS3074	18-1-24		
WM12D07	AACS1602	22-1-24		
WM64D11	AACS1304	18-1-24		

Try fill up
other data
fields

Compound Key

(serves as ONE primary key for
this table)

Database Design Language (DBDL)

- One of the design tools used to represent a data table and its data *fields* or *attributes*.



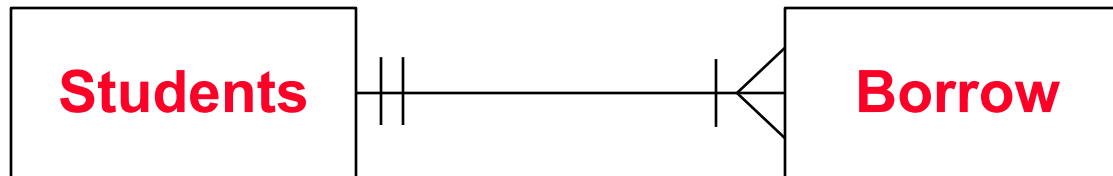
- **DBDL :** STUDENT (Student ID, Name, Address, Advisor No.*)

- **DBDL Components :**
 - **Entity** (Table) Name – in **CAPITAL** letters.
 - **Primary Key** – first field or a combination of fields, to be **underlined**.
 - **Non-Key Fields** – **other** data fields describing other non-key fields.
 - **Foreign Key** (if any) – last field. Indicated with an **asterisk (*)**

Steps in Database Design

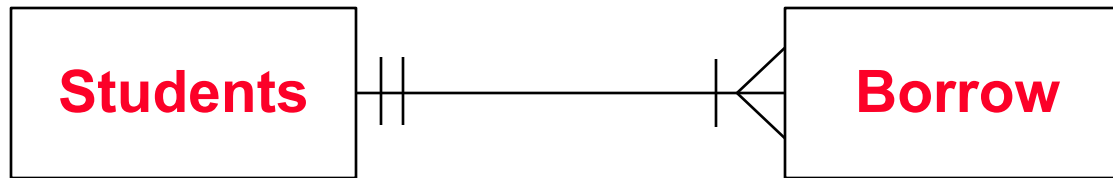
1. Create an **initial ERD**

- Identify 2 entities which are related. This can be carried out using the first 'activity' or 'process'.



2. Assign all data elements to entities

- List **key** and **non-key** data in Database Design Language (DBDL) format.



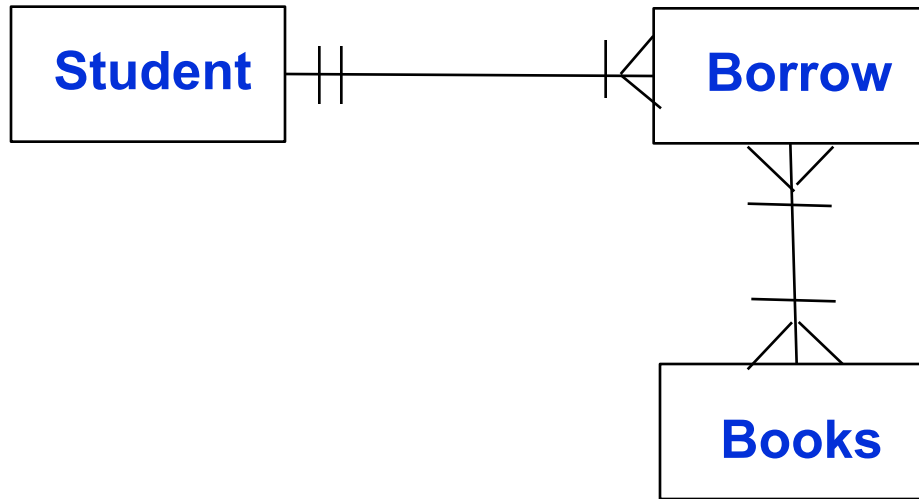
- **Example** (in DBDL format)

STUDENT (Student Number, Name, Address, Handphone)

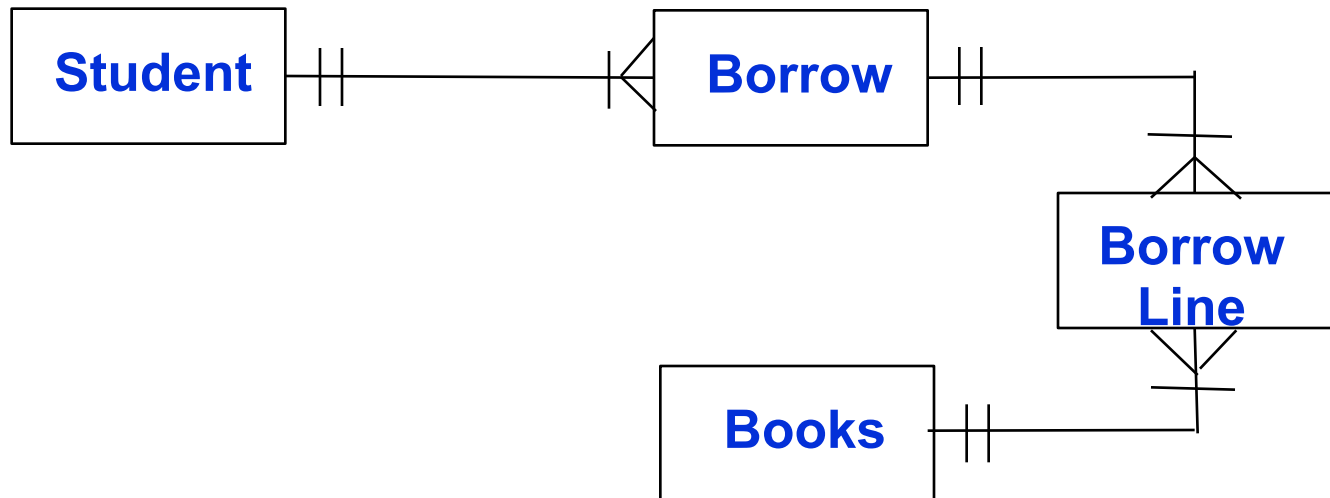
BORROW (Borrow Number, Date, Student ID*)

3. Resolve any many-to-many relationship

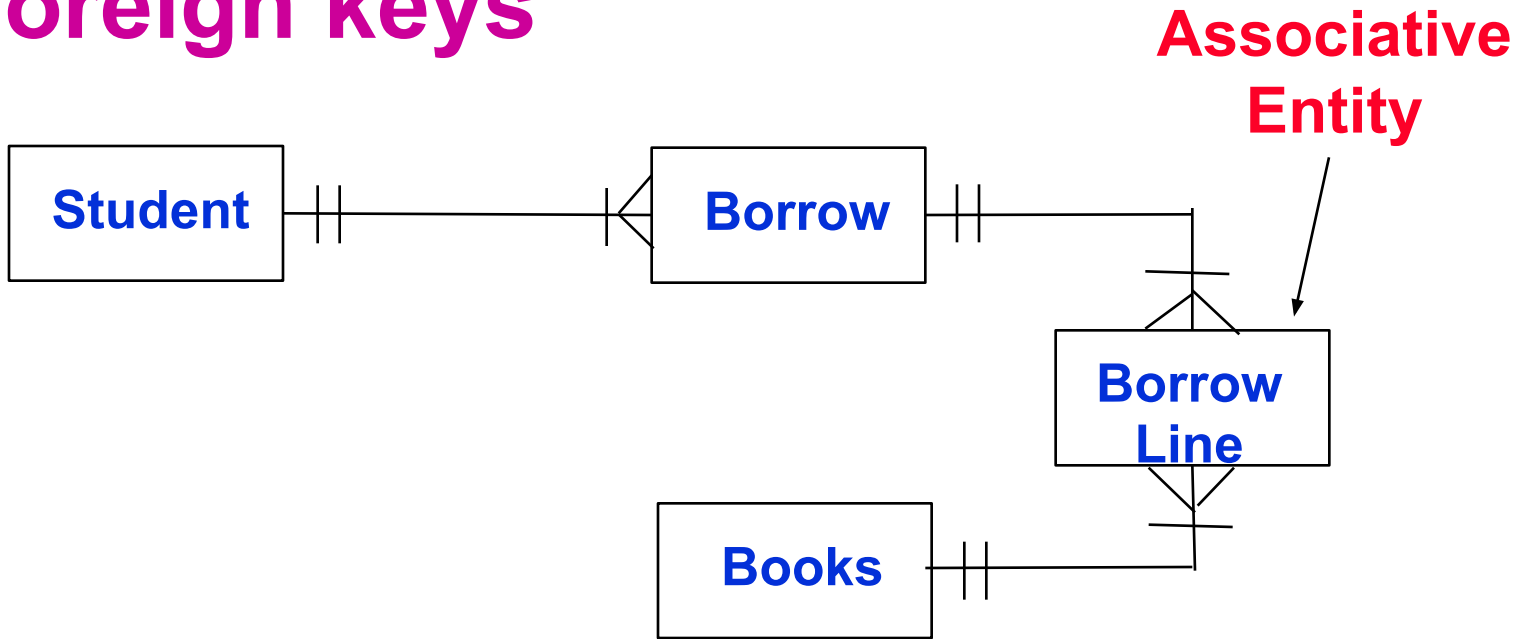
Before



After



4. Relating the Entities by using foreign keys



STUDENT (Student ID, Name, Address, Handphone)

BORROW (Borrow Number, Date, Student ID*)

BOOK (Book Code, Title, Edition, Year-Published)

BORROWLINE (Book Code*, Borrow Number*, Qty)

5. Creating data dictionary entries

- **Data dictionary** – details about **each** data field.
- **Software tools** – eg. Oracle, DB2 can be used.

Data Dictionary

Example of Data Dictionary Format

- **Field name** – the primary name of the data or control item, the data store, or an external entity.
- **Field length** – the number characters
- **Field type** – numeric, alphabetic or alphanumeric
- **Alias** – other names used for the first entry.
- **Field content description** – a notation for representation content.
- **Supplementary information** – other information about: preset values (if known), nullability, restriction or limitation etc

Example of Data Dictionary Tools

The screenshot displays the Microsoft Access interface for a table named 'Client'. The table structure is shown in a grid with columns for Field Name, Data Type, and Description. The 'Field Properties' pane is open, showing the 'General' tab with various field settings. Eight numbered arrows point to specific elements: 1 points to the 'Field Name' column header, 2 points to the 'Data Type' column header, 3 points to the 'Description' column header, 4 points to the 'Field Name' column, 5 points to the 'Field Size' property, 6 points to the 'Format' property, 7 points to the 'Validation Text' property, and 8 points to the 'Required' property.

Field Name	Data Type	Description
Client Number	Text	Client Number (Primary Key)
Name	Text	Client Name
Address	Text	Street Address
City	Text	City
State	Text	State (Two-Character Version)
Zip Code	Text	Zip Code (Five-Character Version)
Telephone Number	Text	Telephone number (8 digits)
Client Type	Text	Client type: EDU - Education, MAN - Manufacturing, SER - Service
Amount Paid	Currency	Amount Paid by Client This Year
Current Due	Currency	Current Due from Client This Period
Trainer Number	Text	Number of Client's Trainer

Field Properties

General | Locks

Field Size	4
Format	0
Input Mask	
Caption	
Default Value	
Validation Rule	
Validation Text	
Required	No
Allow Zero Length	Yes
Indexed	Yes (No Duplicates)
Unicode Compression	Yes
IME Mode	On
IME Sentence Mode	None
Smart Tags	

A field name can be up to 64 characters long, including spaces. Press F1 for help on field names.

Design view. F6 = Switch panes. F1 = Help.

Uses of Data Dictionary

- **During analysis**
 - helps the analyst **organise** information about the data
 - excellent tool for **communicating** with the user
 - serves as a **memory aid**. Any missing information will alert the analyst
- **During design** - as a source of data about data. Used to generate record, file, and data base formats.
- **During implementation** - serves as a common base for all the programmers working on a system.
- **During maintenance** - gives the maintenance programmer a clear idea of how a particular data element is used.

Code Design

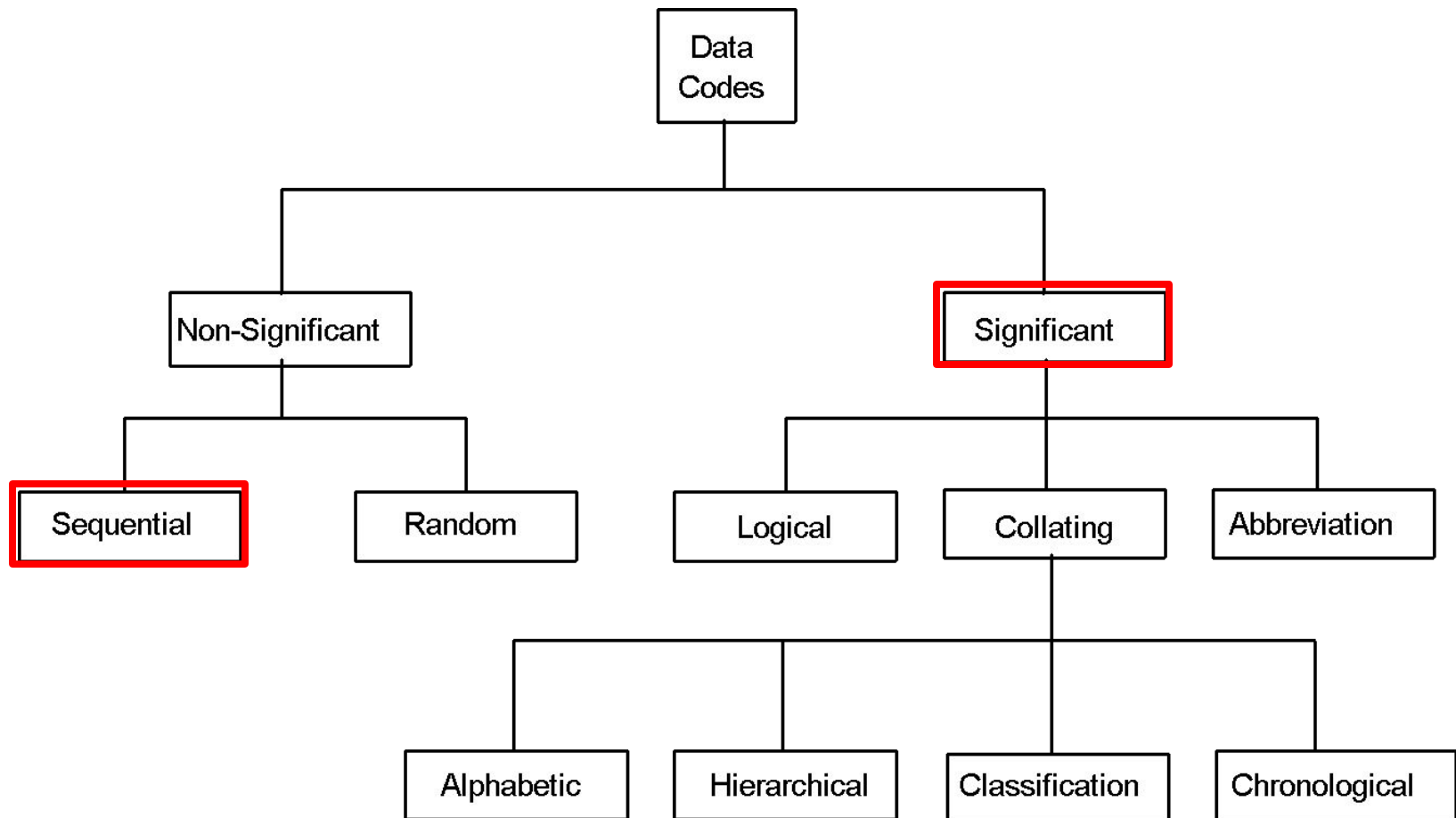
(Primary Key)

Code Design (Primary Key)

- **Meaning** - code is a set of numbers and letters (called a 'primary key') representing an item (a record)
- **Purpose** - to facilitate **unique** identification and retrieval of items of information.

Types of Codes

Example of code : 54-501306-3487



Types of Codes

- **Sequence Codes**

- These are numbers or letters assigned in a specific order. Sequence codes contain no additional information other than an indication of order of entry into the system.
- **Example** : A HRIS issues consecutive employee numbers to identify employees eg 2056 means number 2056th employee.

- **Block Sequence Codes**

- These codes use blocks of numbers for different classifications.
- **Example** : College course number. 100-level courses eg. Chemistry 110 and Mathematics 125, are Year 1 courses, whereas 200s indicate Year 2 courses.

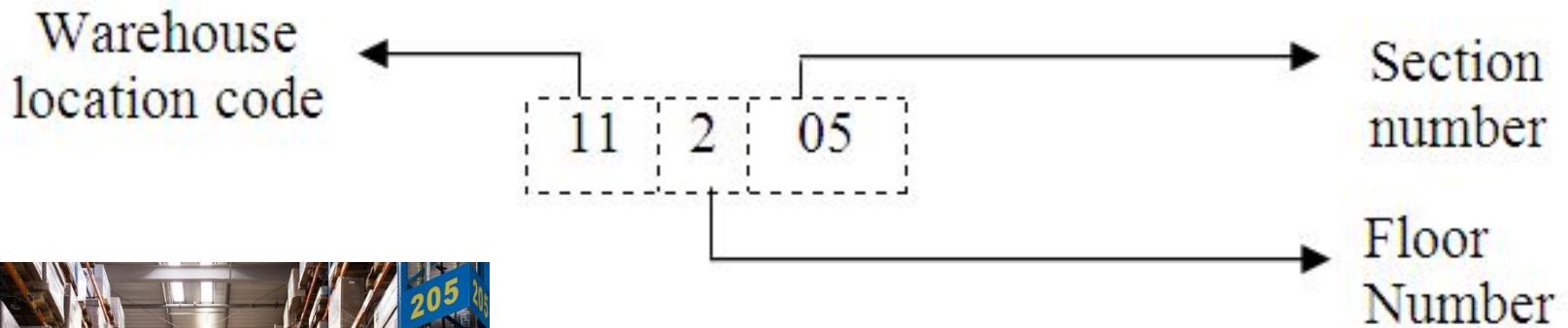
Types of Codes

- Significant **Alphabetic Codes**
 - They use alphabet letters to distinguish one item from another based on a category, an abbreviation, or an easy-to-remember value, called a mnemonic code.
 - **Example :**
 - **SW** for software
 - **HW** for hardware
 - **PL** for plastics

Types of Codes

- Significant **Digit** Codes

- These codes distinguish items by using a series of subgroups of digits.
- **Example** : inventory location codes



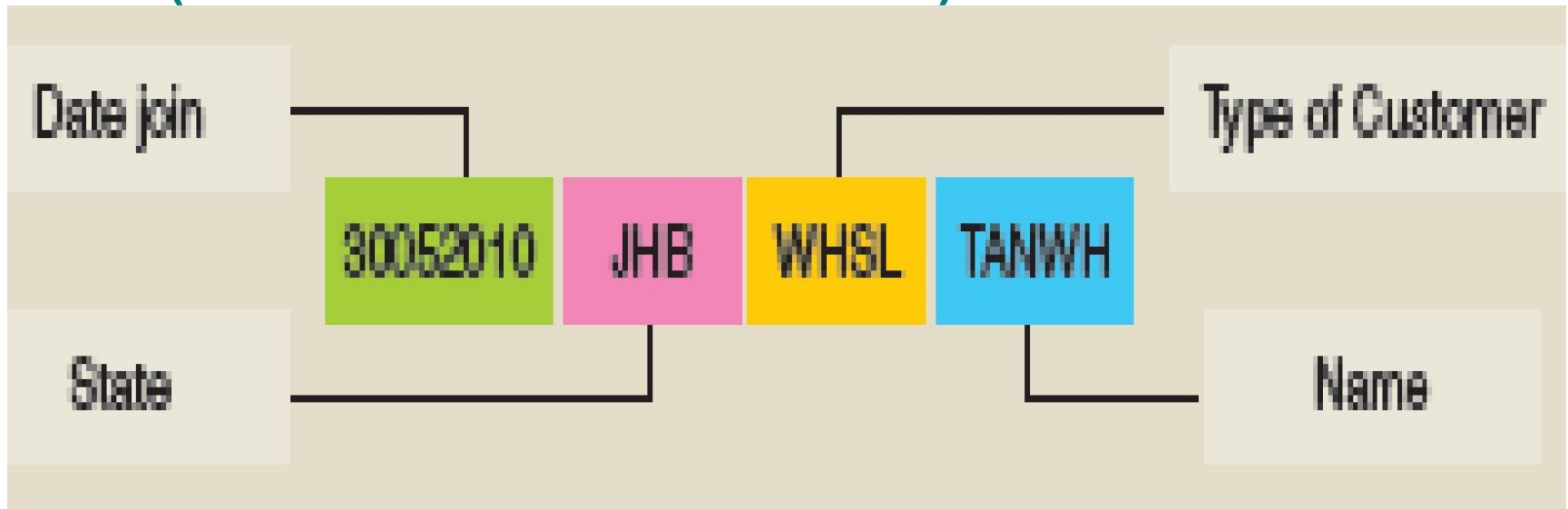
How many Floors can the Code support ?

How many warehouses can the Code support ?

Types of Codes

- **Derivation Codes**

- These codes combine data from item attributes, or characteristics, to build the code.
- **Example** of Customer Number:
(30052010JHBWHSLTANWH)



Students to comment on the major potential problems of using the above Code design

***Additional Info:**

Universal Product Code (UPC)

- **It is a barcode which is uniquely assigned to an inventory item.**
- **The code is used to represent a product in the POS system.**



***Additional Info:**

Quick Response (QR) Code

- It is the trademark for a type of matrix barcode (or two-dimensional code) designed for various industries.
- Recently, the system has become popular due to its fast readability and large storage capacity.



Designing a Code (**Primary Key**)

Generally speaking :

- For **Transaction File** Primary Key – must use *Sequence Code* eg. PO Number, Quotation Number, Invoice Number
- For *complex* **Master File** Primary Key – must use *Significant Code* eg. Stock Code
- For *simple* **Master File** Primary Key – can still use *Sequence Code* eg. Employee No., Customer No.

Exercise for Students

- Suggest and provide example of codes suitable for the following :
 - Customer Number
 - Purchase Order Number
 - Stock Code for clothes

Suggested Answers

- Suggest and provide example of codes suitable for the following :

- Customer Number

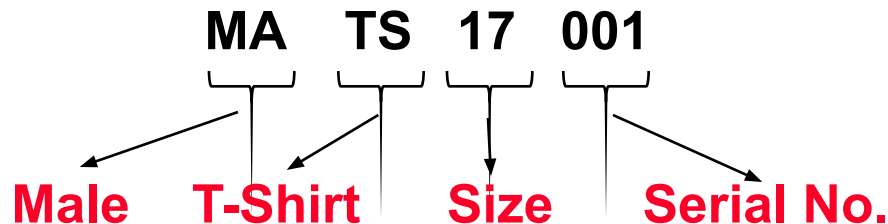
- Use Sequence Code
- 000001 (allow expandability)

- Purchase Order Number

- Use Sequence Code
- 0000000001 (allow expandability)

- Stock Code for clothes

- Use Significant Code



Characteristics of A Good Code

1. **Uniqueness.** The code has only **one** value for a record in the entity. No other records have the same value. The code is able to identify a record uniquely.
2. **Expandability.** The code should be long enough (**sufficient** characters) to allow for possible future growth eg. use more characters, add serial numbers (end), year or alphabets to the code (front).
3. **Conciseness.** The code should be as **short** as possible and still able to represent a record uniquely without sacrificing expandability characteristic.

Characteristics of A Good Code (cont..)

4. **Simplicity.** The code must be simple to remember, understood, interpret and **apply** by the users, particularly workers with little experience eg. use alphabets, M = Medium, L = Large
5. **Stability.** Codes which do not require to be frequently changed or **updated** when record characteristics change.
6. **Flexibility.** The code should be easy to **modify** (eg add new characteristic) to reflect necessary changes in characteristics of a record in the entity.

Exercise for Students

Consider the following relations and express them in a proper **DBDL** format for a **partially** extracted Shipping Database:

- **DESTINATION** (Dest Code, Destination)
- **SHIP** (Ship Id, Model, Weight, Capacity)
- **SCHEDULE** (Dest Code, Ship Id)

- a) Express the above again using a proper DBDL format. Indicate all primary and foreign keys.
- b) Draw an ERD to show the above.

Suggested Answers

a) DBDL

- **DESTINATION** (Dest Code, Destination)
- **SHIP** (Ship Id, Model, Weight, Capacity)
- **SCHEDULE** (Dest Code^{*}, Ship Id^{*} date)

b) ERD

